A.5 Stochastic Centering on the Empirical NTK of Graph Neural Networks

Using a simple grid-graph dataset and 4 layer GIN model, we compute the Fourier spectrum of the NTK. As shown in Fig. 7, we find that shifts to the node features can induce systematic changes to the spectrum.

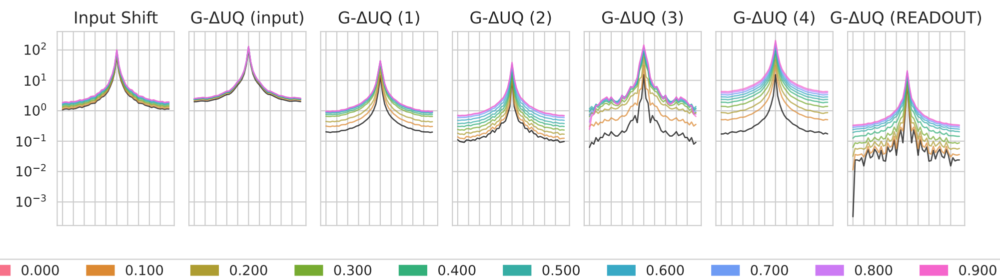

Figure 7: **Stochastic Centering with the empirical GNN NTK.** We find that performing constant shifts at intermediate layers introduces changes to a GNN's NTK. We include a vanilla GNN NTK in black for reference. Further, note the shape of the spectrum should not be compared across subplots as each subplot was created with a different random initialization.

A.6 SIZE-GENERALIZATION DATASET STATISTICS

The statistics for the size generalization experiments (see Sec. 5.1) are provided below in Table 9.

Table 9: **Size Generalization Dataset Statistics:** This table is directly reproduced from (Buffelli et al., 2022), who in turn used statistics from (Yehudai et al., 2021; Bevilacqua et al., 2021).

	NCI1			NCI109		
	ALL	SMALLEST 50%	LARGEST 10%	ALL	SMALLEST 50%	LARGEST 10%
CLASS A	49.95%	62.30%	19.17%	49.62%	62.04%	21.37%
CLASS B	50.04%	37.69%	80.82%	50.37%	37.95%	78.62%
# OF GRAPHS	4110	2157	412	4127	2079	421
AVG GRAPH SIZE	29	20	61	29	20	61

	PROTEINS			DD		
	ALL	SMALLEST 50%	LARGEST 10%	ALL	SMALLEST 50%	LARGEST 10%
CLASS A	59.56%	41.97%	90.17%	58.65%	35.47%	79.66%
CLASS B	40.43%	58.02%	9.82%	41.34%	64.52%	20.33%
# OF GRAPHS	1113	567	112	1178	592	118
AVG GRAPH SIZE	39	15	138	284	144	746

A.7 GOOD BENCHMARK EXPERIMENTAL DETAILS

For our experiments in Sec. 5.2, we utilize the in/out-of-distribution covariate and concept splits provided by Gui et al. (2022). Furthermore, we use the suggested models and architectures provided by their package. In brief, we use GIN models with virtual nodes (except for GOODMotif) for training, and average scores over 3 seeds. When performing stochastic anchoring at a particular layer, we double the hidden representation size for that layer. Subsequent layers retain the original size of the vanilla model.

When performing stochastic anchoring, we use 10 fixed anchors randomly drawn from the in-distribution validation dataset. We also train the G-ΔUQ for an additional 50 epochs to ensure that models are able to converge. Please see our code repository for the full details.

We also include results on additional node classification benchmarks featuring distribution shift in Table 12. In Table 13, we compare models without G-ΔUQ to the use of G-ΔUQ with randomly sampled anchors at the first or second layer.

Dataset	Shift	Train	ID validation	ID test	OOD validation	OOD test	Train	OOD validation	ID validation	ID test	OOD test
		Length									
	covariate	24744	5301	5301	17206	17490					
GOOD-SST2	concept	27270	5843	5843	15142	15944					
		Color									
	covariate	42000	7000	7000	7000	7000					
GOOD-CMNIST	concept	29400	6300	6300	14000	14000					
	no shift	42000	14000	14000	-	-					
		Base					Size				
	covariate	18000	3000	3000	3000	3000	18000	3000	3000	3000	3000
GOOD-Motif	concept	12600	2700	2700	6000	6000	12600	2700	2700	6000	6000
		Word					Degree				
	covariate	9378	1979	1979	3003	3454	8213	1979	1979	3841	3781
GOOD-Cora	concept	7273	1558	1558	3807	5597	7281	1560	1560	3706	5686
		University									
	covariate	244	61	61	125	126					
GOOD-WebKB	concept	282	60	60	106	109					
		Color									
	covariate	420	70	70	70	70					
GOOD-CBAS	concept	140	140	140	140	140					

Table 10: Number of Graphs/Nodes per dataset.

Dataset	Model	# Model layers	Batch Size	# Max Epochs	# Iterations per epoch	Initial LR	Node Feature Dim
GOOD-SST2	GIN-Virtual	3	32	200/100	–	1e-3	768
GOOD-CMNIST	GIN-Virtual	5	128	500	–	1e-3	3
GOOD-Motif	GIN	3	32	200	–	1e-3	4
GOOD-Cora	GCN	3	4096	100	10	1e-3	8710
GOOD-WebKB	GCN	3	4096	100	10	1e-3/5e-3	1703
GOOD-CBAS	GCN	3	1000	200	10	3e-3	8

Table 11: Model and hyperparameters for GOOD datasets.